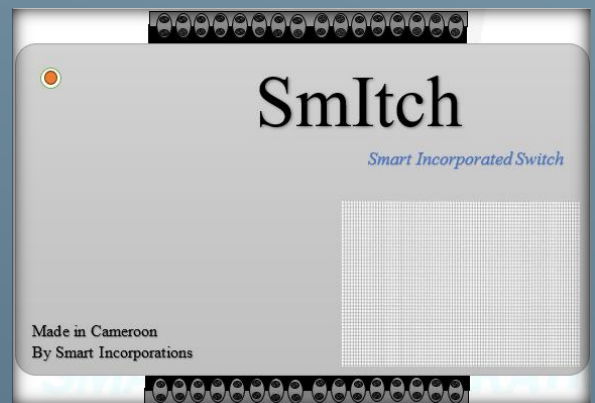


SMART INCORPORATIONs

Exploring technology beyond limits



V₁

SMART INCORPORATED SWITCH

SmItch Installation Guide V1

Location: Bafoussam

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Foreword

Smart Incorporations offers a wide range of Smart Home Devices, and we realize that with all the standards, rules, listings and codes, the what, when, where, why and how of smart home devices can get complicated. The following information is provided to aid in the proper use of Smart Incorporations smart home device (SmItch) and all its capabilities.

This guide is written as a general guide for people working with SmItch app and device as well as for those who are simply interested in learning more about the products, standards and applications.

About this manual



Please read, understand, and follow all safety information contained in this manual before installing or using this equipment.

Save this installation manual and keep it near the equipment. If you are unsure about any part of this installation or of the potential hazards discussed, please contact your supervisor or supplier immediately.

All specifications and descriptions contained in this document are verified to be accurate at the time of printing. However, because continuous improvement is a goal at Smart Incorporations, we reserve the right to make product modifications at any time.

The images provided in this document are for demonstration purposes only. Depending on product version and market region, details may appear slightly different.

ERRORS OR OMISSIONS

To communicate any inaccuracies or omissions in this manual, send an email to smartincorporations@gmail.com



ELECTRONIC DEVICE: DO NOT THROW AWAY

Proper Disposal of device is required. Refer to your local codes for disposal requirements.

Made in Cameroon

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Technical Support

If you have questions about this device, its use, or operation, please contact your dealer or call the Smart Incorporation Technical Service department at +237 671449466 / 696906980, or send a mail to us at smartincorporations@gmail.com.

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SmItch Hardware

I. Safety

Obey the safety instructions when you install, operate and do maintenance on this device. Obey these instructions to prevent physical injury or death, and damage to SmItch or to connected equipment.

Safety and warnings



SmItch is used for high voltage switching. Contact with live electrical parts can result in serious injuries or even death! Live parts are all metal parts belonging directly to one of the circuits or wires connecting to them. All other visible metal parts and wiring may also be live if a fault exists. Before starting any work on the contactors, always comply with the following safety rules:

- Disconnect on all sides
- Secure to prevent switching back on
- Clearly identify the working area Besides the main power circuits, also disconnect additional and auxiliary circuits

In the case of a failure of the equipment;

- Don't use it anymore
- immediately contact the manufacturer.



This symbol in conjunction with the signal word "WARNING" indicates a potentially dangerous situation. Failure to observe the related safety note may cause personnel injury or death or equipment damage.



This symbol in conjunction with the signal word "NOTE" indicates operator tips, particularly useful or important information for the use of the product. This symbol and wording do not indicate a dangerous situation.



This symbol indicates a compulsory action: "Reading the instruction manual/booklet before starting work or before operating equipment or machinery.



Recycle.



Do not dispose of this in ordinary trash.

II. Intended Use

SmItch is to be used in domestic and commercial electrical installations.

III. Receiving

Upon reception of the equipment, make an immediate inspection for any damages which may have occurred during shipment. When SmItch is received it should be unpacked sufficiently to inspect for concealed damage and to determine that the shipment is complete and correct. If any damage is found, it should be noted prior to accepting the shipment if possible. Check for things such as, damage to the container, water seepage or moisture on the SmItch device.

If any parts are damaged or missing, file a claim immediately with the carrier and notify the Smart Incorporations sales office. Try to give as much information as possible when filing a claim.

IV. Handling & storage

SmItch should be handled with care to avoid damage. This device should be kept in an upright position and moved with suitable equipment. If SmItch is to be stored over any length of time, the packing material should be restored for protection during that period. Where conditions permit, the packing should be left intact until the device is ready for installation. Store the SmItch in a dry and dust free area. It should not be left outdoors or in adverse conditions. If unprotected, corrosion, rust and insulation deterioration may occur.

V. Safety Considerations



Please consider the following safety issues before beginning the installation of SmItch. Although we have compiled this list of common safety considerations, it should not be considered as complete. It is not intended to take the place of your good judgment, training, and experience.

i. Personal Safety Equipment and Clothing

Only certified Electrical Technicians are authorized to install this device. Personal safety equipment and clothing including high visibility vests, hard hats, gloves, safety shoes, safety glasses are just some of the items available to you.

Choose the right equipment for the job. If you are unsure of which safety equipment is recommended or appropriate for the job, ask your supervisor or foreman.

ii. Work Zone Traffic Control

Proper control of persons is important during the installation of this device. It is important that you use appropriate work zone traffic control techniques, equipment, and procedures. If you are unsure of which procedures are recommended or appropriate for the job, ask your supervisor or foreman.

VI. Safety in installation and maintenance

Obey the local regulations and standards and the instructions in this manual, when you do work on the:

- SmItch
- Input and output cabling

i. General safety



Do not repair a malfunctioning unit. Contact your supplier, the local SI representative or an Authorized Service Center for instructions.

- Make sure that dust or water does not go into SmItch during installation.
- Make sure that air circulates around SmItch inlets

ii. Electrical safety



Only qualified electricians can install and connect the SmItch.

To ensure the safety and optimal functionality of the device, we strongly emphasize the need to carefully follow the prescribed configurations and instructions outlined in the accompanying documentation. Any deviation from the recommended guidelines may lead to irreversible consequences, including the risk of fire or irreparable damage.

1. Electrical safety rules

- Obey all electrical safety regulations of the installation country or region.
- Be aware that SmItch is connected to two voltage sources:
 - The device operates on DC power for all slots designated as P (Inputs). Under no circumstances should AC power be supplied to the P input slot, While the output for P slots is AC.
 - The device operates on AC power for all slots designated as S (I/O).
- Do not open SmItch cover.
- Before you do work on SmItch, isolate the SmItch from all power sources.
- Do not disconnect SmItch when it's under load. Use sealing caps to prevent water.
- The mounting surface should be vertical (level; ± 1 mm). If the mounting surface is not level, adjust with spacers or shims.
- In wiring the main circuit, wires should be given sufficient length to be flexible for service.

- Avoid touching the surface of the SmItch with soiled, wet or dirty hands. You can damage the silicon finish on the contact. If the contacts become dirty, simply clean it with a non-tracking cleaner such as rubbing alcohol.
- Use accepted and recognized safety precautions to prevent the risk of electric shock or electrocution.
- This device should be used only for its intended purpose.
- keep out of reach of children.

DO NOT install the Smitch in the following areas:

- Areas with high salt content, such as the marine environment. It will deteriorate metal parts, causing the parts to fail or the unit to leak water.
- Areas filled with mineral oil or containing a large amount of splashed oil or steam, such as a kitchen. It will deteriorate plastic parts, causing the parts to fail or the unit to leak water.
- Areas that generate substances that adversely affect the equipment, such as sulfuric gas, chlorine gas, acid, or alkali. It will cause the copper pipes and brazed joints to corrode, which can cause refrigerant leakage.
- Areas that can cause combustible gas to leak, which contains suspended carbon-fibre or flammable dust, or volatile inflammable such as paint thinner or gasoline.
- Environments where precipitation or humidity are above 95%.

You're understanding and adherence to these directives are paramount to maintaining the integrity and longevity of the device. Should you require any further clarifications or assistance, please consult the provided documentation or contact our technical support team immediately.

You're understanding and adherence to these directives are paramount to maintaining the integrity and longevity of this device.

2. Grounding safety (PE)



Obey these instructions to prevent injury or death, or damage to the equipment.

- Obey all of the electrical safety regulations of the installation country.
- Connect the protective earth (PE) for all sockets and associated equipment externally to ensure personal safety and device protection.
- Make sure that the grounding connections meet the safety regulations.
- Do not ground the P (input slots) carrying DC.

VII. Safe start-up and operation



Obey these instructions to prevent injury or death, or damage to the equipment.

1. Before you start SmItch

- Make sure SmItch is protected Miniature Circuit Breaker
- Check all P slots and S slots to make sure AC is not supplied to a DC terminal.

VIII. Target audience

This manual is written for persons who plan the installation, and install, commission, use and do maintenance on this home automation device. Read the manual before you do work on SmItch. You are expected to know the fundamentals of electricity, wiring, electrical components and electrical schematic symbols.

IX. SmItch models

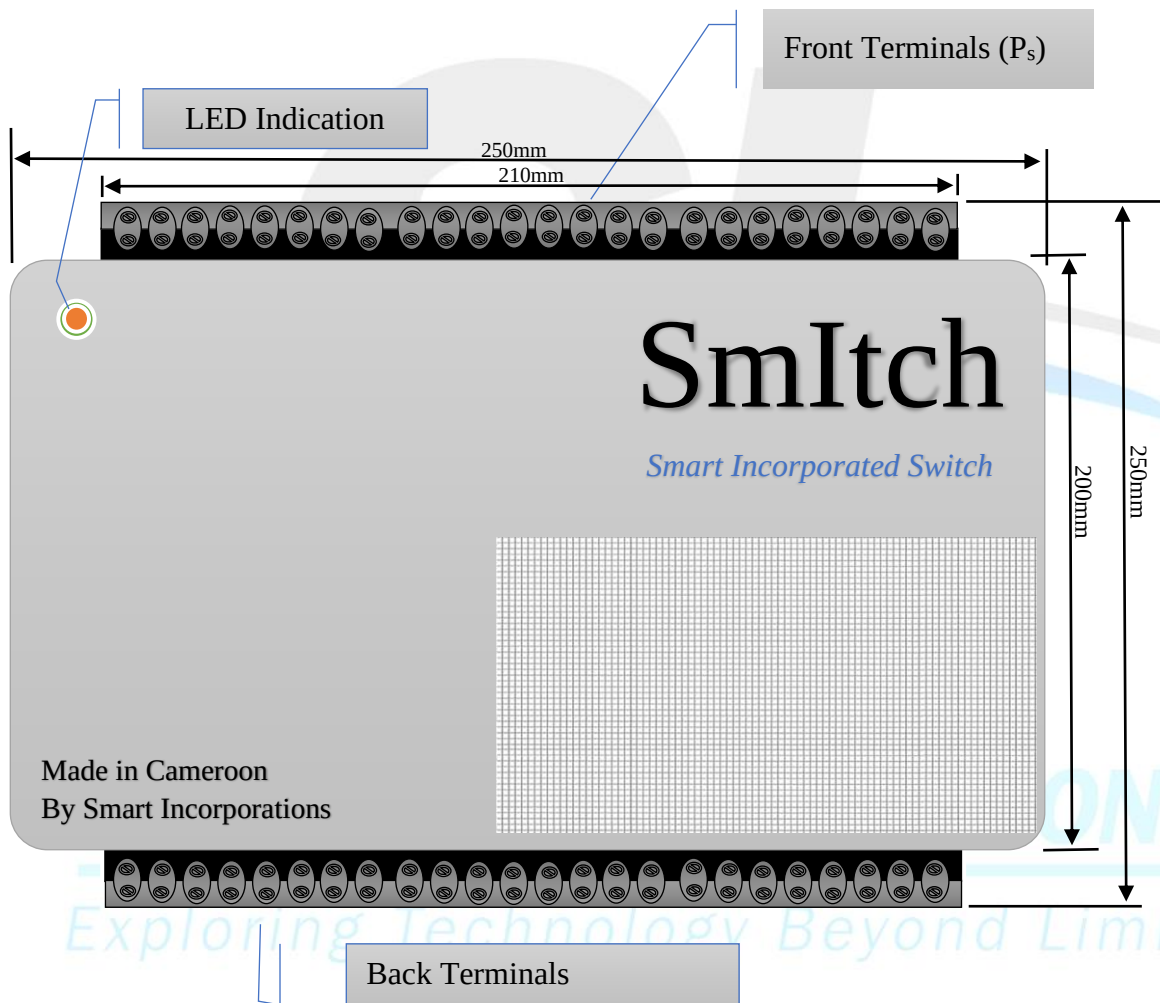
<i>Model</i>	Description
<i>SmItch Mini</i>	SmItch Mini has 16 I/O slots. Mini has 8 P slots Inputs all in DC and 8 S slots all in AC
<i>SmItch Mega</i>	SmItch Mega has 32 I/O slots. Mega has 16 P slots Inputs all in DC and 16 S slots all in AC



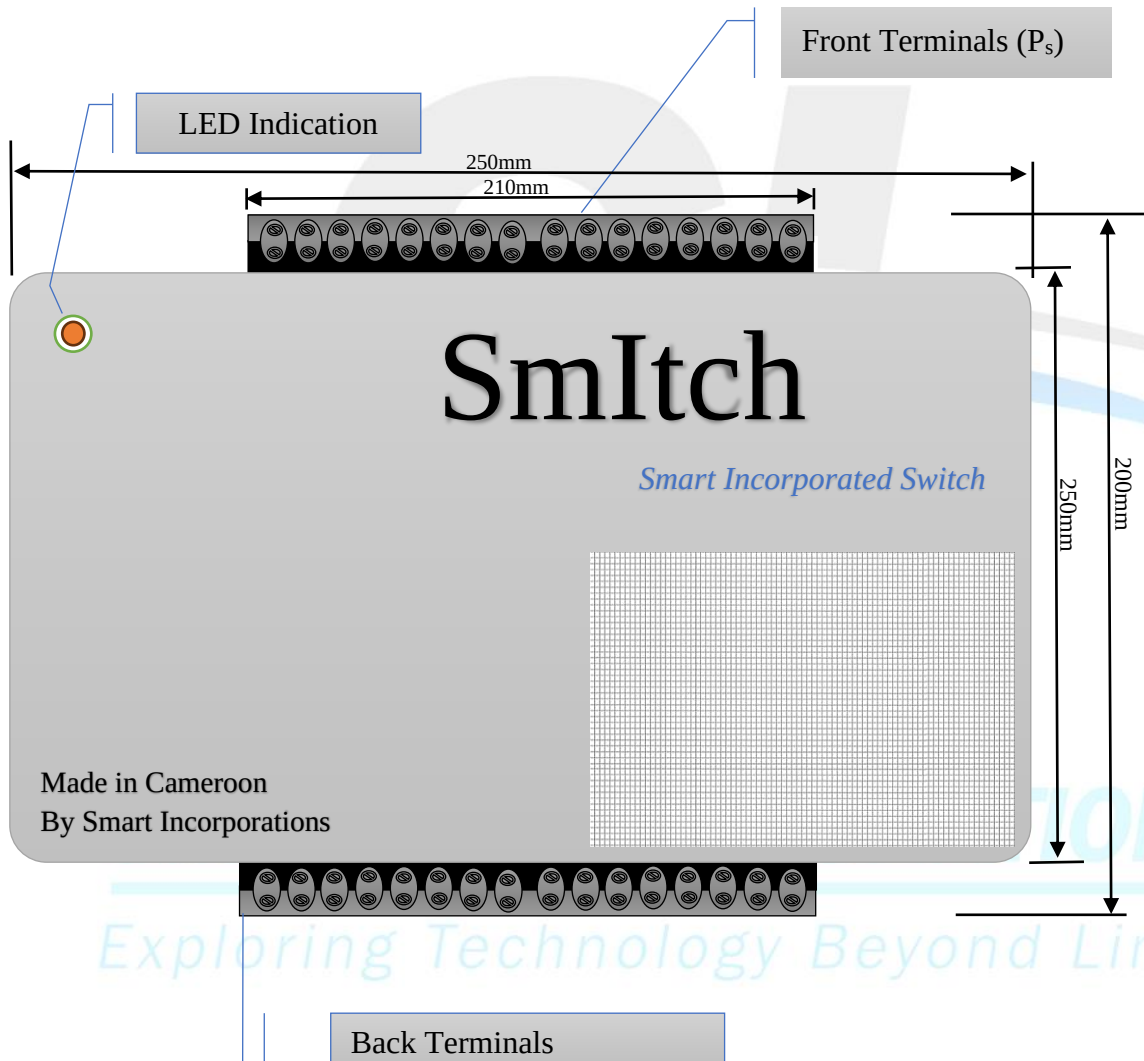
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Wiring of SmItch

I. SmItch Mega Diagram

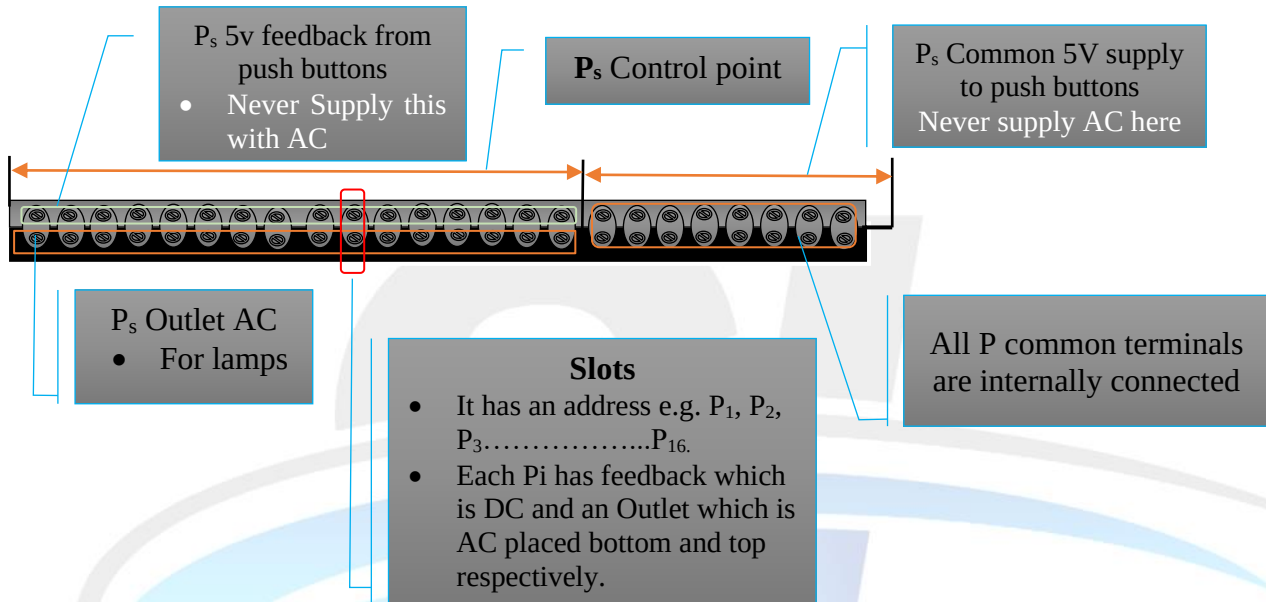


II. SmItch Mini

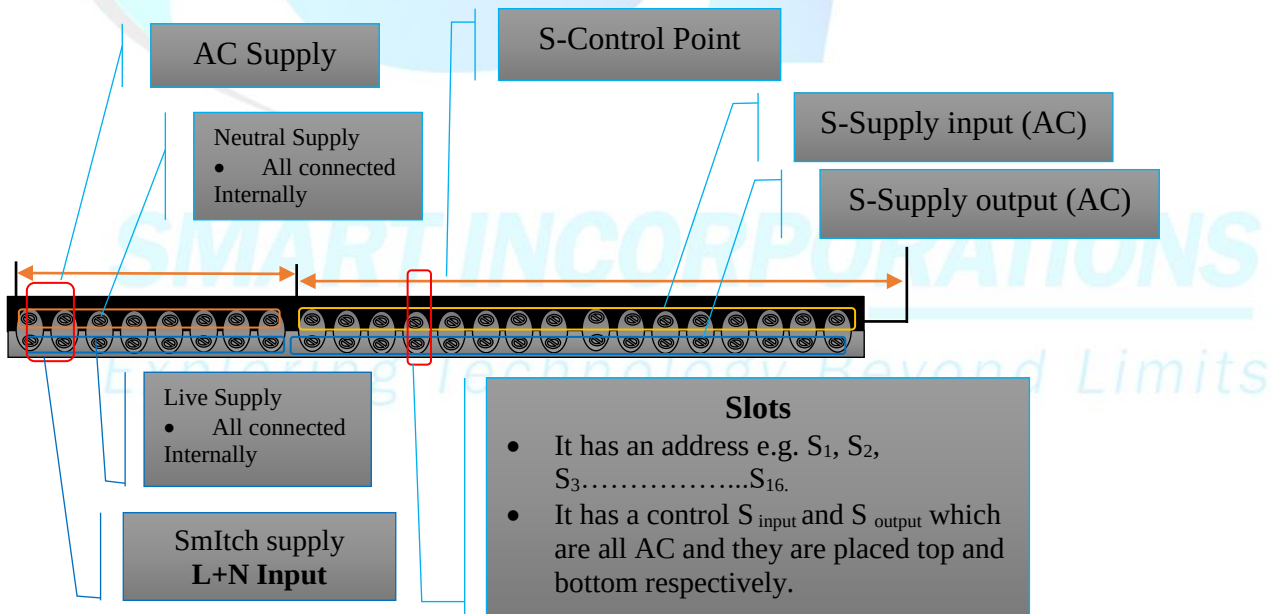


III. SmItch Mega Terminals

- Control Terminals (P_s)



- Control terminals for S



NB: For Mega and Mini

P_s - Starts from the left to right i.e., $P_1, P_2, P_3, \dots, P_n$

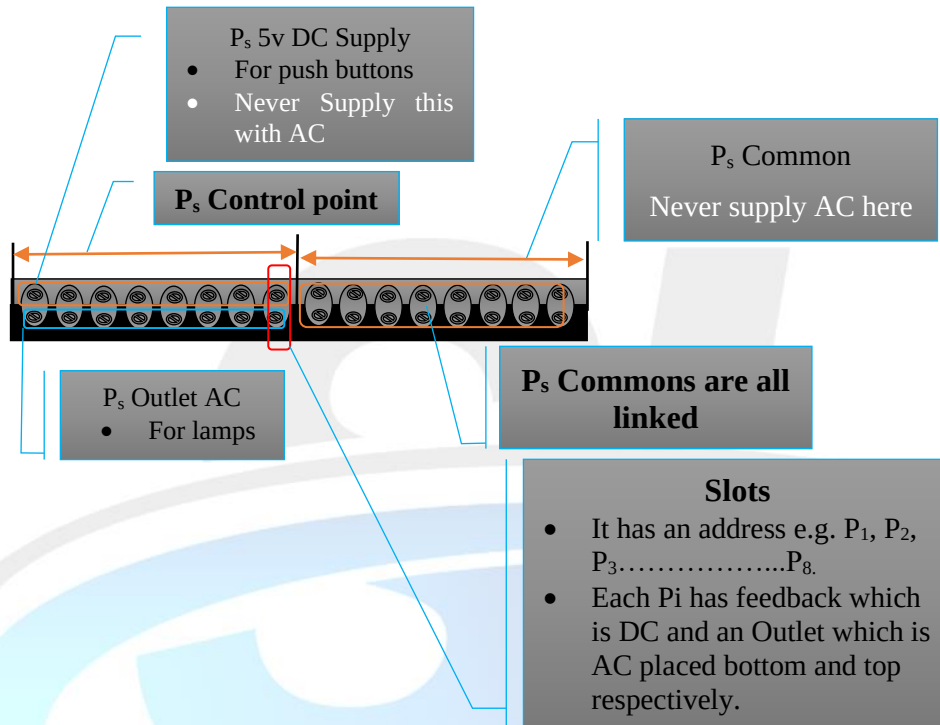
$P_1 \longrightarrow P_n$

S_s - Starts from the right to the left i.e., $S_1, S_2, S_3, \dots, S_n$

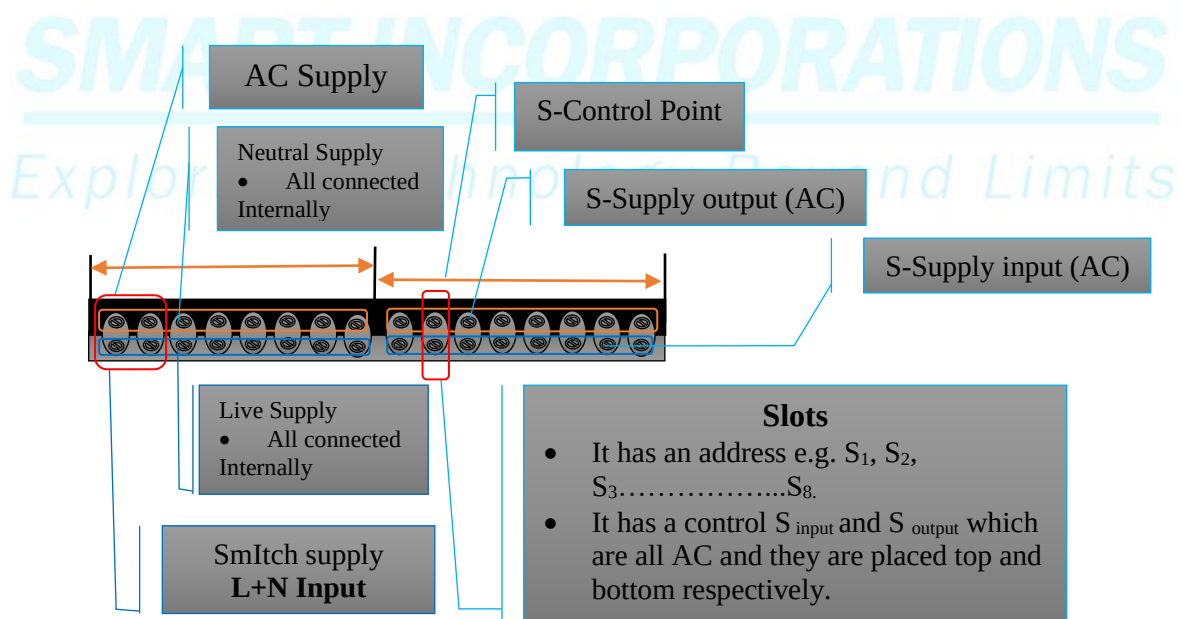
$S_n \longleftarrow S_1$

IV. SmItch Mini Terminals

• Control Terminals (P_s)



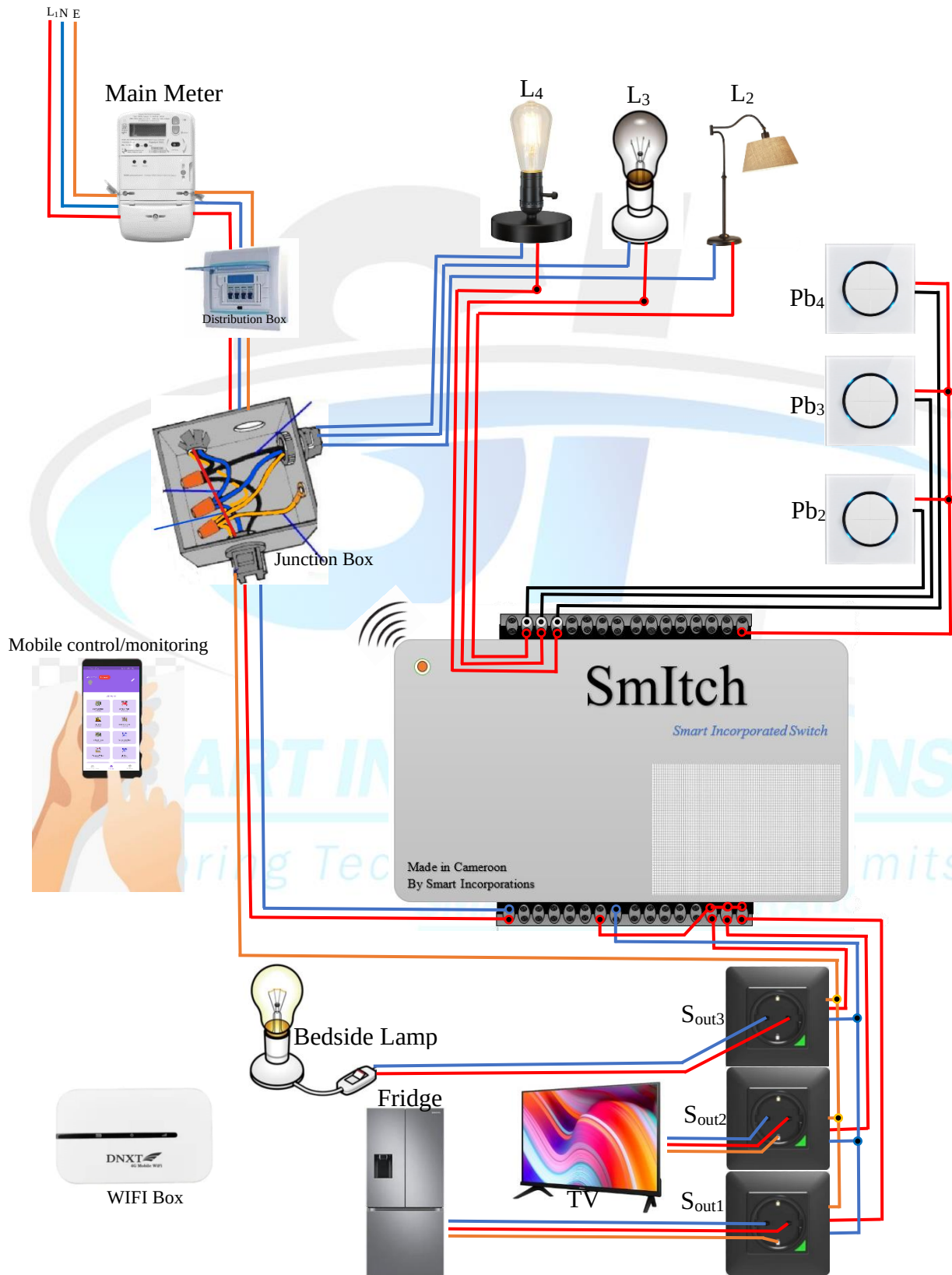
• Control Terminals S



SmItch Connection

I. Layout/wiring

- Exo:1



II. SmItch LED indicators

Operation	STATUS	DESCRIPTION
Power On	Blinking Red	When the SmItch device is powered on, it will automatically turn on and initiate a network search.
Network Connection	Steady Green.	Once the SmItch device successfully finds and establishes a connection with the network, this indicates that the device is connected to the network. NB: here it can take up to two minutes to subscribe online if that isn't done then it will automatically switch to either steady red or blinking red
Modem Connected	Blinking Green	When SmItch is connected with the modem and there is data connection, this indicates that data is actively being transmitted between the Smitch device and the modem.
Modem Connected	Steady Red	When SmItch is connected with the modem and there is no data connection, this indicates that data is actively not being transmitted between the SmItch device and the modem.

NB: The above diagram is a brief and explanatory layout on how SmItch is installed

III. Installation area



WARNING! Do not install the SmItch:

- Near corrosive, explosive or flammable materials
- Where it is subject to impacts or vibration
- In a dusty place
- In an area with conductive dust
- In an area where snow or sand piles
- Near to a heat source. Heat decreases the power output and the service life of the unit.

i. Tools required

- Screw driver set,
- Hand Drill
- Tester
- Multimeter

IV. Configuration

- Configuration for Exo:1

SmItch Config										
						Date completed:				
Project: Smitch Configuration			Location: ----- -----			Dwg: N/A		Contract N°: --		
Estimator: Smart Incorporations						Location: Bafoussam				
SmItch Type						Mini				
Project Summary										
INSTALLATION										
	N°	Items	Port	Description	Control device		X-tics	Qty	Price	
	00	//	//	//	Pbs	Sock			Unit	Total
-----Test' s Room-----	01	Center Lights	P ₂	L ₂	02	--	NO	01	--	--
	02	Wall Lamps	P ₃	L ₃	03	--	NO	01	--	--
	03	Table Lamp	P ₄	L ₄	04	--	NO	01	--	--
	03	Bedside Lamp	S ₁	S _{out1}	--	01	220-230v AC	01	--	--
	04	TV outlet	S ₂	S _{out2}	--	02	220-230v AC	01	--	--
	05	Fridge outlet	S ₃	S _{out3}	--	03	220-230v AC	01	--	--
	NB	Buzzer is for emergencies' only and also, only applicable on the base SmItch if any is mounted in your location with another	P ₁	Bz	n	----	-----	01	-----	-----
	NB	For all Ps you can used cables of 0.5mm to 1.5mm	----	-----	----	----	-----	----	-----	-----
	NB	For all Ss use normal 2.5mm cables								
	07	-----	S _n P _n	-----	----	----	-----	----	-----	-----

Troubleshooting

I. Faults

Fault	Cause	What to do
AC overcurrent	- AC-side short-circuit - Sudden AC-side voltage drop	If the P and S outlet AC-side circuit breaker on the electrical distribution board is tripped, an electrician must examine your electrical system. Report the behaviour to your system supplier. Check your supply line
SmItch Can't connect online	- No Internet access - No WIFI	Activate internet connection Redo WIFI configuration
App can't monitor and control online	- No internet access - SmItch offline	Check home WIFI connection Check phone mobile connection
Can't change WIFI network	You have set WIFI name and Password as that of the company	Check your phone's hotspot name and password Redo the WIFI configuration
Physical devices (switch/socket) don't work	- App disables physical - Relay damaged	Enable physical device via mobile Check if that device isn't scheduled Visit Smart Incorporation personnel for check up
WIFI configuration isn't connecting	Network connection	When you disconnect your old WIFI actuate any physical switch

II. Regular tests/checks

All of the maintenance activities that may be carried out on SmItch by skilled personnel are listed below.

Component	Tests/checks	Measures
Voltage connection cables/busbars	Check for: • dirt • damage or wear • corrosion • damaged cables, • cable lugs • damaged insulation • damaged busbars • kinked or crushed cables • loose or missing fastening elements • correct tightening torque of the terminal screws	In case of faults: • remove any existing dirt • immediately replace damaged cables or cable lugs • immediately replace damaged busbars • remove corrosion and replace corroded parts • tighten loose fastening elements and replace missing fastening elements immediately • tighten the terminal screws to the rated torque
SmItch/base plate fastening	Check for: • loose or missing fastening elements	In case of faults: • tighten loose fastening elements and replace missing fastening elements immediately • tighten a loose contactor immediately
Main contacts S and P slots	Thereafter, the main contacts are accessible and can be checked for: • damage or wear • traces of combustion residue (slight soot deposits permitted) A certain level of experience is required to assess the state of the contacts. Even after a few load switching cycles, to a lay user the contacts appear worn and the surroundings fouled. Replacement of the main contacts is only necessary if the minimum height of the main contact	In case of faults: • if the contact material on any of the fixed contacts or on the contact bridge has burnt off to less than 0.3 - 0.4 mm minimum height

	tips is less then 0.3 - 0.4 mm (new condition approx. 1.2 - 1.4 mm).	
Relay coil	Check for: • dirt • damage or heavy wear • damage due to operation with impermissible, significantly too high coil voltage	In case of faults: • if damaged or heavily worn, replace the complete relay coil

III. Decommissioning

i. To remove SmItch from service



To remove SmItch from service:

1. Switch OFF the system.
2. Electrically isolate SmItch
3. Disconnect the P and S slots from the switches/lamps and circuit breakers.
4. Make sure that the disconnected cables do not cause danger.
5. Place them in an identifiable manner so as to easy replace
6. Remove the locking screws from the back of SmItch if it was wall mounted but if its in the distribution box remove it gently.
7. To remove SmItch from the wall-mounting plate, tilt the bottom of SmItch outwards and lift SmItch upwards.
8. Recycle SmItch responsibly.

ii. Recycling



Do not dispose SmItch with normal waste.



At the end of its operational life, return SmItch to any recycling facility which receives and handles electronic goods. Alternatively, the product can be returned to Smart Incorporations for discount for the next SmItch purchase depending on the reason it's damaged. Please contact your local Smart Incorporations representative for assistance.

Technical data

I. Manufacturer

Address	Smart Incorporations Cameroon _Bafoussam
Article number/Ordering data	

II. Smltch Properties

No.	Component	Specification
<i>Electrical parameters</i>		
	Tested according to	
	Protection class	IP40
	Ambient temperature	-45°C to 80°C
	Operating temperature	-40°C to 85°C
	Humidity range	0% to 70%
	Pollution degree	Negligeable noise
	Rated operational voltage U_e range	100V to 250V
	Rated control voltage U_p	0V to 600V
	Rated insulation voltage U_i	0V to 1KV
	Rated impulse withstands voltage U_{imp}	275V
	Rated current I_n for P_s	10A
	Rated current I_n for S	30A
	Idle power consumption for mega	324mW
	Idle power consumption for mini	324mW
	Max power consumption for Mega	21.124W
	Max power consumption for Mini	10.724W
	Switching capacity for mega	45KW
	Switching capacity for mini	22.5KW
	Max switching capacity for mega	90KW
	Max switching capacity for mini	42KW
	Coil switching time	10uS
	Push bottom respond time	100ms

Arc quenching		Yes	
Protective features		Fuse	
Ripple and noise		Handled	
Cooling method		Fan	
Dimensions		250mm by 250mm	
Wiring			
Control circuit	Hard wire	1.~.5 mm ²	
	Flexible wire	0.5~1 mm ²	
Power circuit	Hard wire	2.5~6 mm ²	
	Flexible wire	1.5~4 mm ²	
Mechanical service life		100 years	
Pollution degree		Negligeable noise	
Installation class		-	
Weight		2.7Kg	
Conformity		HIGH	
Colour		White	

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III. DC Input side

- For all P input slots

No.	Component	Specification
	Number of inputs for mega	16
	Number of inputs for mini	8
	Input current I_n	454uA
	Permitted voltage range U_n	0V-5V DC
	Max line resistance	9K Ohm
	Max power consumed on single button click	2.27mW
	Max power consumption for mega	36.32mW
	Max power consumption for mini	18.16mW
	Breakdown voltage	6.7V DC
	Input voltage	5V DC
	Max input voltage	6.6V DC
	Switch type	Tactile push button

IV. AC Input side

- For all S input slots

No.	Component	Specification
	Number of slots for mega	16
	Number of slots for mini	8
	Input voltage	220V
	Input current	30A
	Input power	6.6KW
	Max input voltage	600V
	Max input current	30A
	Max input power	18KW
	Breakdown voltage	>1KV
	Breakdown current	>30A

V. Connectivity

No. Component Specification

<i>SmItch</i>				<i>SmItch mobile application</i>	Specification	
System initialization time		9000ms to 30000ms		Command time	500ms to 1000ms	
Data processing time		3ms		Response time	1500ms to 3000ms	
Online response time		200ms to 300ms		Loading time	500ms to 5000ms	
Data retrieving time		1ms		Control range	Worldwide	
State update time		280ms to 1900ms				
New WIFI search timeout		5000ms per search				
Resubscription time		2 Minutes				
WIFI connection time		1ms				

- **Remark:** This time is based on average network

NB: Before any installation, Smart Incorporations must first Configure your SmItch device before any installation is carried out.

This can only be done by an administrator at Smart Incorporations.